Florentin GUTH

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Employment

Incoming Assistant Professor — 2027–
RLE & EECS, Massachusetts Institute of Technology — Cambridge, USA

Research Fellow — 2023–
Center for Computational Neuroscience, Flatiron Institute — New York, USA

Faculty Fellow — 2023–2026
Center for Data Science, New York University — New York, USA

Research Intern — Summer 2022
Center for Computational Neuroscience, Flatiron Institute — New York, USA

Research Intern — Summer 2018
Facebook AI Research — New York, USA and Paris, France

Education

Ph.D. in Computer Science — 2019–2023
École Normale Supérieure — Paris, France
Thesis title: *Towards a Mathematical Understanding of Deep CNNs*. Advisor: Prof. Stéphane Mallat.

M.Sc. in Maths, CV and ML (MVA) — 2018–2019
École Normale Supérieure Paris-Saclay — Paris, France
Advanced courses in mathematics, computer vision and machine learning.

ENS Diploma — 2016–2019
École Normale Supérieure — Paris, France
Bachelor- and Master-level courses in computer science and mathematics.

Classe Préparatoire — 2014–2016
Lycée du Parc — Lyon, France
Intensive undergraduate courses in mathematics, physics, and computer science.

Publications

- Yimiao Yu, **Florentin Guth**. *Localizing Transfer Between Memorization Tasks*. In **Neural Artifacts workshop, NeurIPS, 2026**.
- Jeevesh Attri, **Florentin Guth**. *Memorization in image classification: test set behavior and scaling laws*. In **Neural Artifacts workshop, NeurIPS, 2026**.
- Nicolas Zilberstein, Santiago Segarra, Eero Simoncelli, **Florentin Guth**. *Learning normalized energy models for linear inverse problems*. In **ICML 2026**.

- Jamie Simon, Daniel Kunin, Alexander Atanasov, Enric Boix-Adserà, Blake Bordelon, Jeremy Cohen, Nikhil Ghosh, **Florentin Guth**, Arthur Jacot, Mason Kamb, Dhruva Karkada, Eric Michaud, Berkan Ottlik, Joseph Turnbull. *There will be a scientific theory of deep learning*. Preprint.
- Guy Ohayon, Pierre-Étienne Fiquet, **Florentin Guth**, Jona Ballé, Eero Simoncelli. *Learning a distance measure from the information-geometry of data*. In **ICLR 2026**.
- **Florentin Guth**, Zahra Kadkhodaie, Eero Simoncelli. *Learning normalized image densities via dual score matching*. In **NeurIPS, 2025**.
- Louis Thiry, **Florentin Guth**. *Classification-denoising networks*. Preprint.
- **Florentin Guth**, Brice Ménard. *On the universality of neural encodings in CNNs*. In **Re-Align workshop, ICLR, 2024**.
- Zahra Kadkhodaie, **Florentin Guth**, Eero Simoncelli, Stéphane Mallat. *Generalization in diffusion models arises from geometry-adaptive harmonic representations*. In **ICLR, 2024**. ***Outstanding Paper Award***.
- **Florentin Guth**, Brice Ménard, Gaspar Rochette, Stéphane Mallat. *A Rainbow in Deep Network Black Boxes*. In **JMLR, 2024**.
- **Florentin Guth***, Etienne Lempereur*, Joan Bruna, Stéphane Mallat. *Conditionally Strongly Log-Concave Generative Models*. In **ICML, 2023**.
- Zahra Kadkhodaie, **Florentin Guth**, Stéphane Mallat, Eero Simoncelli. *Learning Multi-Scale Local Conditional Probability Models of Images*. In **ICLR, 2023**. ***"Top 25%"***.
- **Florentin Guth**, Simon Coste, Valentin De Bortoli, Stéphane Mallat. *Wavelet Score-Based Generative Modeling*. In **NeurIPS, 2022**.
- **Florentin Guth**, John Zarka, and Stéphane Mallat. *Phase Collapse in Neural Networks*. In **ICLR, 2022**.
- John Zarka, **Florentin Guth**, and Stéphane Mallat. *Separation and Concentration in Deep Networks*. In **ICLR, 2021**.

Invited Talks

Redwood Center Seminar, UCBerkeley	Berkeley, USA	10/16/26
Microsoft Research New England	Boston, USA	08/11/26
SIERRA, INRIA	Paris, France	06/29/26
ENS CSD Seminar	Paris, France	06/24/26
MIT EECS Seminar	Boston, USA	03/19/26
UCBerkeley EECS Seminar	Berkeley, USA	04/12/26
Cornell SDS Seminar	Ithaca, USA	02/18/26
Weissman Lab, Stanford	Stanford, USA	11/20/25
Columbia ML and AI Seminar	New York City, USA	11/07/25
Yale SDS Seminar	New Haven, USA	10/27/25
Brown CS Theory Seminar	Providence, USA	10/24/25
Johns Hopkins MINDS-CIS Seminar	Baltimore, USA	10/14/25
Princeton Language and Intelligence Seminar	Princeton, USA	10/02/25
Kempner Diffusion Seminar	Cambridge, USA	09/29/25
Michaeli Lab, Technion	Online	09/13/25
UniReps Speaker Series	Online	07/31/25
LemanTH Workshop, EPFL	Lausanne, Switzerland	05/16/25
LeCun Lab, NYU	New York City, USA	04/02/25
Pruna AI	Paris, France	01/23/25
CFM	Paris, France	09/12/24
Maths and ML Conf., IMSA	Miami, USA	06/26/24
Youth in High Dimensions, ICTP	Trieste, Italy	05/23/24
TAG-DS Seminar	Online	01/30/24
DeepMath Conf.	Baltimore, USA	11/17/23
Keynote and Tutorial, CCN Conf.	Oxford, UK	08/26/23
MALGA Seminar	Genova, Italy	07/17/23
Rough Path Interest Group	Online	10/05/22
4th IMA Conf. on Big Data	Oxford, UK	09/20/22

Bonner Lab, JHU	Baltimore, USA	07/12/22
MLIA, LIP6, Sorbonne U.	Paris, France	12/09/21
SIERRA, INRIA	Paris, France	11/25/21
3IA Doctoral Workshop	Toulouse, France	11/22/21
DL Theory Seminar, FAIR	Online	11/12/21

Teaching

CDS Instructor

2023–2026

Center for Data Science, NYU — New York, USA

Taught DS-GA 1014: Computational Linear Algebra and Optimization" (Master's level) for Fall 2023–2025. Also gave invited lectures in Prof Rajesh Ranganath's (graduate) and Prof Joan Bruna's (undergraduate) classes.

Great feedback from students (average satisfaction 4/5).

Maths Teaching Assistant

2020–2023

CPES, PSL — Paris, France

Exercise sessions for undergraduate students.

200h of experience. Great feedback from students.

AI Teacher

2021–2023

Ipesup — Paris, France

I designed and teach a 24-hour-long introductory course on Artificial Intelligence to high-school students. Topics include the history, applications, limitations, ethical concerns, computational and mathematical aspects of AI, as well as practical programming sessions.

Great feedback from students (average satisfaction of 9/10).

Maths and CS Teacher

2019–2021

Optimal Sup-Spé — Paris, France

Classes and exercise sessions for undergraduate students.

More than 200h of experience. Great feedback from students.

Awards & Honors

- **NeurIPS Top AC**, 2026.
- **ICLR Outstanding Paper Award**, 2024.
- **NeurIPS Top Reviewer**, 2023 & 2025.
- **ENS PhD Fellowship** (*CDSN*), 2020–2023.
- **Bridge France Champion** (*under 18, by pairs*), 2013.

Students

- **Jinghan Lei** (Spring 2026–, Master's research project): Rate-distortion flows via successive refinement of information.
- **Yimiao Yu & Jeevesh Attri** (Spring–Summer 2026, Master's research project): Scaling laws of memorization and transfer learning in image classification tasks.
- **Nicolas Zilberstein** (Summer–Winter 2025, Graduate internship): Solving inverse problems with anisotropic diffusion energy models.
- **Sixto Cerna Lopez** (Spring 2022, Undergraduate internship): Implementation of machine learning algorithms for real-estate price estimation.

Community

Workshop on Scientific Methods for Understanding Deep Learning

ICLR 2026 — Rio de Janeiro, Brazil — 04/26/26
NeurIPS 2024 — Vancouver, Canada — 12/15/24

Co-lead organizer. Both editions received 100 submissions, estimated 700 unique participants (300 peak), > 500 followers on X/Twitter . Website and videos of 1st and 2nd editions.

Conferences & Journals

Area chair for NeurIPS (2026–), ICLR (2027–).
Reviewer for NeurIPS (2023–2025), ICLR (2024–2025), JMLR (2024), ACHA (2025–), Nature Machine Intelligence (2026–).

Service

NYU Paris SSAC Committee (2023–2026), NYU CDS Master Admissions (2024–2026), NYU CDS Master Scholarships (2025–2026).

Outreach

My Personal Experience Doing Research

Collège de France — Paris, France — 03/22/21

10-minute-long talk about my research topic, why I love research and my background, given to 100 high-school students from underprivileged areas.

How to Create a Programming Language?

Savoirs Pour Tous, ENS — Paris, France — 11/08/18

1-hour-long wide-audience talk on compilation, aimed at high-school students. Video and slides (in French).